

Operators

↓
Addition, comparing values or
assigning value to a
variable

7 types of operators

① Arithmetic operators

② Comparison operators

↪ return
True Boolean
 results
false

③ Logical Operator

AND

↓

Return

true if

both operands are True

OR

↪ Return

true if

atleast one
operand is True

NOT

↪ Opposite

Boolean

value of

the

operand

$$\begin{cases} x = 5 \\ y = 10 \\ z = 15 \end{cases}$$

$$\boxed{x < y} \quad \underline{\text{And}} \quad \boxed{y < z} \longrightarrow T$$

$$\boxed{x < y} \quad \underline{\text{Or}} \quad \boxed{x > z} \longrightarrow T$$

$$\text{NOT} (x < y) \longrightarrow F$$

④ Bitwise Operator Binary Number

& (Bitwise AND)

$$x = 10$$

$$y = 6$$

→ 1 0 1 0

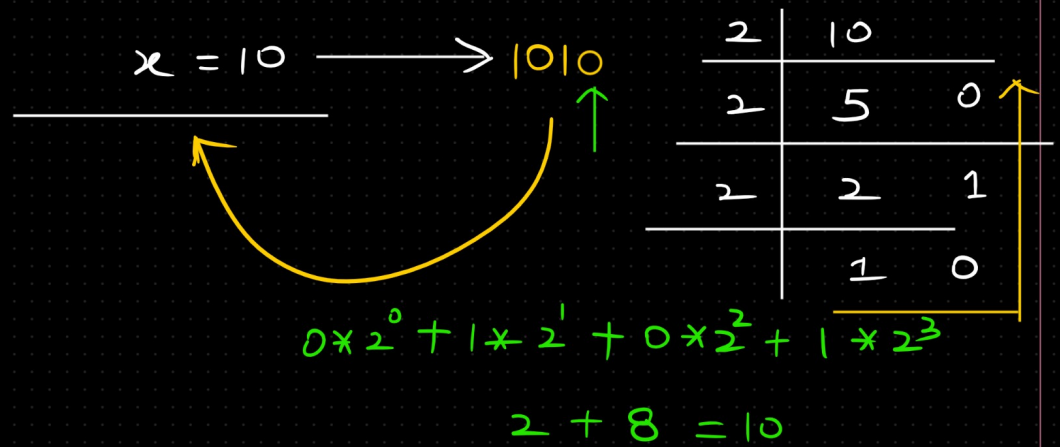
→ 0 1 1 0

1010

0110

0010 ← 2

Binary Number (0 & 1)



✓

| $\longrightarrow$ Bitwise OR

✓
^

$\longrightarrow$ Bitwise XOR

`print(x ^ y)`

bits different

1 0 1 0 ✓
0 1 1 0 ✓

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \longrightarrow 1$

1 1 0 0 $\longleftarrow$ 12 ✓

BITWISE NOT (~)

$\longrightarrow$ Internal working

Left Shift

times shift
operation
 $x \times 2^1$
 $10 \times 2^1 = 20$

$x = 10 \rightarrow$

 $x \ll 1$

$\begin{array}{cccc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{array}$
 $\hookrightarrow 4 + 16$

$= 20$

$\downarrow$
20

$x \ll 2$

 $10 \times 2^2 = 40$
 $\hookrightarrow 40$

$\begin{array}{ccccc} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{array}$
 $\hookrightarrow 8 + 32 = 40$

$x \ll 3$

$\hookrightarrow 10 \times 2^3 = 80$

Right shift

$$x = 10 \longrightarrow \underline{1} \underline{0} \underline{1} 0$$

$$x \gg 2 \longrightarrow \underline{0} \underline{1} \underline{0} \underline{1}$$

5

$$x \div 2^3 \rightarrow \underline{\underline{\# \text{ shift}}}$$

$$10 \div 2 = \underline{\underline{5}}$$

Assignment Operator

Membership Operators

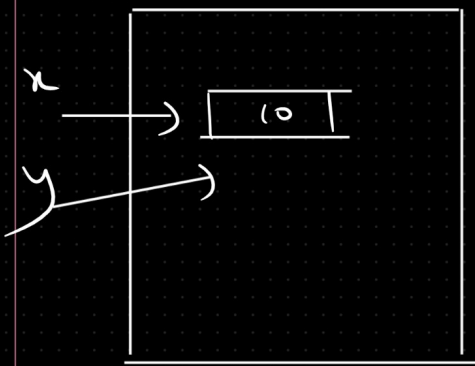
① in _____

② not in _____

Identity Operator

① is _____

② is not _____



$\begin{cases} x = 10 \\ y = 10 \end{cases}$

$x \text{ is } y \longrightarrow \text{true}$

Data type

Strings $\longrightarrow$ Data Structure
 $\searrow$ List functionality
 $\searrow$ sequence of char

"Python"

Programming $\longrightarrow$

0	1	2	3	4	5	(Positive index)
P	y	t	h	<u>o</u>	<u>n</u>	
-6	-5	-4	-3	-2	-1	

$\text{Programming}[4] = 'o'$

$\text{Programming}[-1] = 'n'$

Reverse string

$\text{string}[:, -1] \Rightarrow$

$\text{string}[-4:] \Rightarrow$ old!

$\text{string}[:, -5] \Rightarrow$ Hello, W

String $\Rightarrow$

0	1	2	3	4	5	6	7	8	9	10	11	12
H	e	l	l	o	,		w	o	r	l	d	!
-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1

String[start : end : step_size] $\text{len(string)} = 13$

String[1:4] $\Rightarrow$ ell

String[1:8:2] $\Rightarrow$ el, w

String[:5] $\Rightarrow$ Hello

String[7:] $\Rightarrow$ World!

Mutable $\rightarrow$ change

immutable $\rightarrow$ Can't change

String[: :-1] $\Rightarrow$

String[: :-2] $\Rightarrow$

$\text{for}(i=0; i < n; i++)$	$\text{for}(i=n-1; i \geq 0; i--)$
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